

# Ensemble

M5 Retail Demand Forecasting — Problem Statement 11

Phase 8 — a real MAE gain, no RMSE gain

NPN AIA Hackathon — St. Joseph's College of Engineering

\*Phase 8 — blending the two objectives that won on different metrics. Generated 2026-08-14.\*

**Terms.** **RMSE** — average error with big misses punished far more heavily; lower is better. **MAE** — the plain average error. **Leakage** — letting the model see information that would not have existed when the forecast was really made. **Tweedie** — a loss function for data that is never negative and mostly zero. **Objective / loss function** — the definition of "wrong" that the model tries to minimise. **Inner window** — an earlier 28-day period we tune on, so the real scoring window stays untouched.

Every number here comes from an experiment that actually ran. Failed experiments are reported alongside successful ones — in this campaign most of them failed, and that is the finding.

## Why these two

Phase 6 produced a clean split: Tweedie is best on RMSE, L1 is best on MAE. Blending them tests whether the trade-off can be improved rather than merely slid along.

## Weight selection (inner window only)

| Weight on Tweedie | Inner RMSE | Inner MAE |
|-------------------|------------|-----------|
| 0.0               | 2.1826     | 0.9293    |
| 0.1               | 2.1579     | 0.9347    |
| 0.2               | 2.1365     | 0.9409    |
| 0.3               | 2.1184     | 0.9480    |
| 0.4               | 2.1037     | 0.9557    |
| 0.5               | 2.0925     | 0.9641    |
| 0.6               | 2.0848     | 0.9733    |
| 0.7               | 2.0807     | 0.9831    |
| 0.8               | 2.0802     | 0.9937    |
| 0.9               | 2.0833     | 1.0051    |
| 1.0               | 2.0899     | 1.0173    |

The inner window preferred **0.80 Tweedie / 0.20 L1**, where it beat pure Tweedie by about 0.010 RMSE.

## Applied once to the primary window

|               | RMSE    | MAE           |
|---------------|---------|---------------|
| Tweedie alone | 2.1210  | 1.0319        |
| Ensemble      | 2.1272  | <b>1.0128</b> |
| Change        | +0.0062 | -0.0191       |

The RMSE gain did not transfer — again — but **the MAE improvement is substantial and real: -0.0191**, taking us to 1.0128 against the team's reported 1.0869.

## Decision

Not selected as the final model, because the project's primary metric is RMSE and the ensemble is slightly worse there. But it is **the best model we have for MAE**, it is cheap (no extra training — it blends two models we already have), and it should be the submission if the hackathon turns out to score MAE.